

Fundamentals of Linear Control:

A Concise Approach

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Chapter 4: Feedback Analysis

linearcontrol.info/fundamentals

Recall from Chapter 3

Time-invariant linear system in the time-domain

Impulse response

$$g(t), \quad t \geq 0$$

Convolution

$$y(t) = \int_{0^-}^{\infty} g(t - \tau) u(\tau) d\tau$$

Time-invariant linear system in the frequency-domain

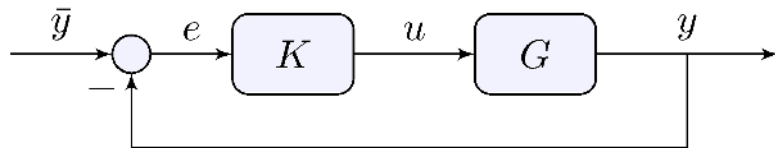
Transfer-function

$$G(s) = \mathcal{L}\{g(t)\}$$

Product

$$Y(s) = G(s) U(s), \quad Y(s) = \mathcal{L}\{y(t)\}, \quad U(s) = \mathcal{L}\{u(t)\}$$

4.1 Tracking, Sensitivity, and Integral Control



Loop relations

$$Y(s) = G(s) U(s), \quad U(s) = K(s) E(s), \quad E(s) = \bar{Y}(s) - Y(s)$$

Closed-loop sensitivity transfer-function

$$E(s) = S(s) \bar{Y}(s), \quad S(s) = \frac{1}{1 + G(s) K(s)}$$

Short-hand notation

$$e = S \bar{y}, \quad S = \frac{1}{1 + G K}, \quad \text{etc}$$

Example: car model with proportional feedback

Open-loop transfer-function

$$G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}}$$

Closed-loop sensitivity transfer-functions

$$S(s) = \frac{1}{1 + K \frac{\frac{p}{m}}{s + \frac{b}{m}}} = \frac{s + \frac{b}{m}}{s + \frac{b}{m} + K \frac{p}{m}}$$

Closed-loop stability

$$-\left(\frac{b}{m} + \frac{p}{m}K\right) < 0 \quad \implies \quad K > -\frac{b}{p}$$

Tracking error

Reference input

$$\bar{y}(t) = \bar{y} \cos(\omega t + \phi), \quad t \geq 0$$

Steady-state from frequency response

$$e_{\text{ss}}(t) = \bar{y} |S(j\omega)| \cos(\omega t + \phi + \angle S(j\omega)) \quad \implies \quad |e_{\text{ss}}(t)| \leq |S(j\omega)| |\bar{y}|$$

e.g. car model with a step reference

$$|e_{\text{ss}}(t)| \leq |S(0)| |\bar{y}| = \frac{\frac{b}{m}}{\frac{b}{m} + \frac{p}{m} K} |\bar{y}| = \frac{1}{1 + \frac{p}{b} K} |\bar{y}|$$

Asymptotic tracking

Definition

Error converges to zero for large time

$$\lim_{t \rightarrow \infty} e(t) = 0$$

Stability of S is not enough. It is necessary that

$$E = S \bar{Y}$$

Poles of $\bar{Y}$ must be cancelled by zeros of S !

Lemma 4.1

Let $S = (1 + G K)^{-1}$ be asymptotically stable and

$$\bar{y}(t) = \bar{y} \cos(\omega t + \phi)$$

If $G K$ has a pole at $s = j\omega$ then $\lim_{t \rightarrow \infty} e(t) = 0$

Proof: S has a zero at $s = j\omega$

Example: toilet bowl model

Open-loop transfer-function

$$G(s) = \frac{1}{sA}$$

Closed-loop sensitivity transfer-function

$$S(s) = \frac{1}{1 + G(s)K} = \frac{1}{1 + \frac{K}{sA}} = \frac{s}{s + \frac{K}{A}}$$

Asymptotically stable for $K > 0$

S has a zero at zero: asymptotic tracking of step (constant) reference inputs

Example: car model with integral feedback

Open-loop transfer-function

$$G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}}$$

Integral control

$$K(s) = \frac{K}{s}$$

Closed-loop sensitivity transfer-functions

$$S(s) = \frac{1}{1 + \frac{K}{s} \frac{\frac{p}{m}}{s + \frac{b}{m}}} = \frac{s \left(s + \frac{b}{m} \right)}{s^2 + \frac{b}{m} s + K \frac{p}{m}}$$

Asymptotically stable for $K > 0$

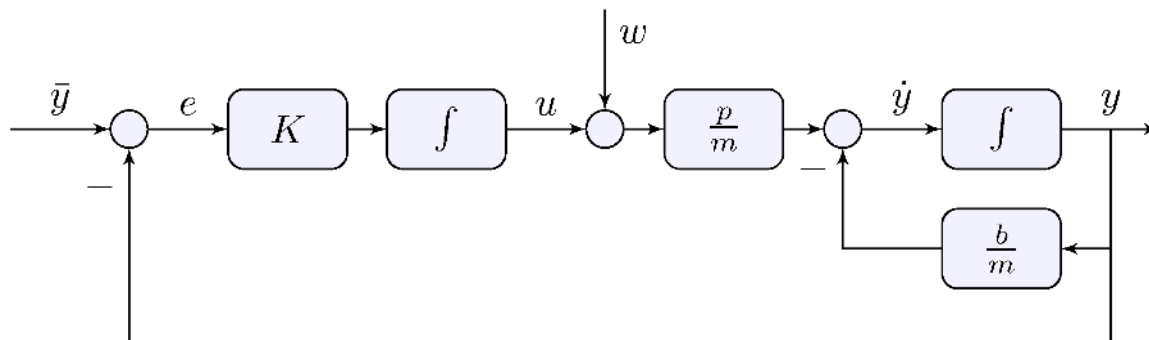
S has a zero at zero: asymptotic tracking of step (constant) reference inputs

Example: car model with integral feedback

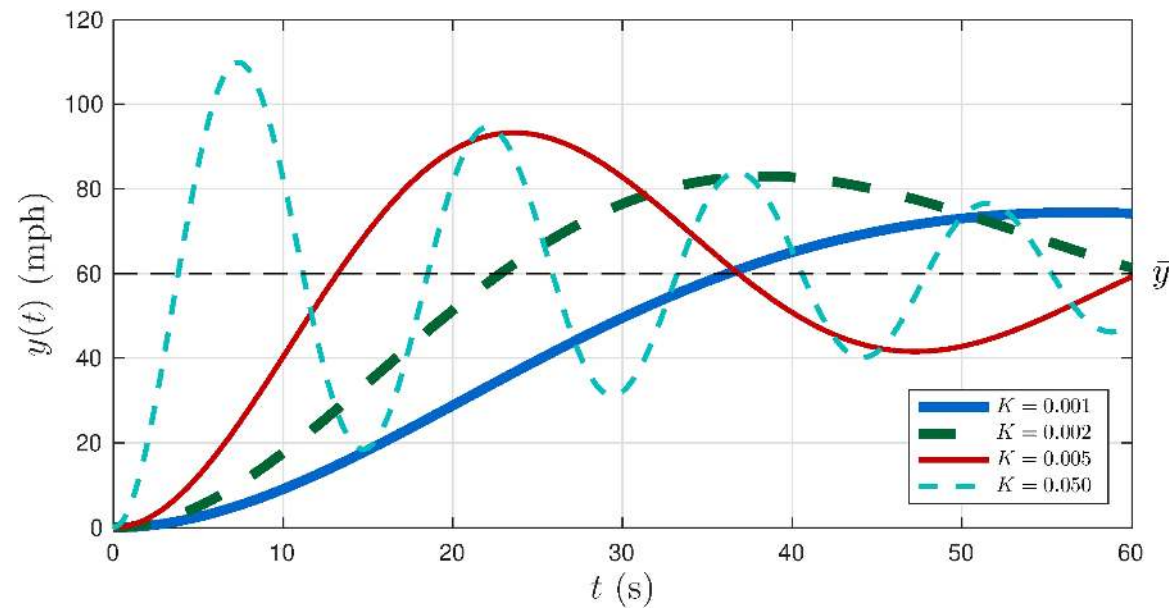
Controller

$$U(s) = \frac{K}{s} E(s) \quad \Longrightarrow \quad u(t) = u(0) + K \int_0^t e(\tau) d\tau$$

System + Controller block-diagram



Example: car model with integral feedback



Transient response has oscillations!

4.2 Stability and Transient Response

Some facts

- The zeros of $1 + GK$ are the poles of S
- The poles of the product GK are the zeros of S
- When G and K are rational the order of S is the order of the product GK

Example: rational transfer-function with no pole-zero cancellations

$$G = \frac{N_G}{D_G}, \quad K = \frac{N_K}{D_K}, \quad S = \frac{1}{1 + \frac{N_G}{D_G} \frac{N_K}{D_K}} = \frac{D_G D_K}{D_G D_K + N_G N_K}$$

For tracking

- Add poles to K so they become zeros of S
- More complex K leads to more complex S
- Watch out for stability and oscillations, e.g. car model with integral feedback

Example: car model with proportional plus integral feedback

Proportional plus integral control

$$G(s) = \frac{p/m}{s + b/m}, \quad K(s) = K_p + s^{-1}K_i$$

Closed-loop sensitivity transfer-functions

$$S(s) = \frac{1}{1 + G(s)K(s)} = \frac{s \left(s + \frac{b}{m} \right)}{s^2 + \left(\frac{b}{m} + \frac{p}{m}K_p \right)s + \frac{p}{m}K_i}$$

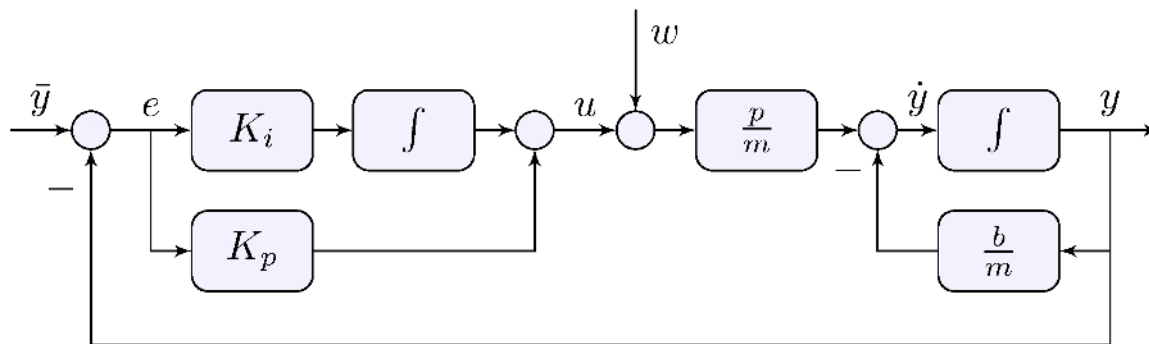
No complex-poles (oscillations) if

$$0 < 4\frac{p}{m}K_i \leq \left(\frac{b}{m} + \frac{p}{m}K_p \right)^2$$

Example: car model with proportional plus integral feedback Controller

$$U(s) = \left(K_p + \frac{K_i}{s} \right) E(s) \quad \Longrightarrow \quad u(t) = u(0) + K_p e(t) + K_i \int_0^t e(\tau) d\tau$$

System + Controller block-diagram



Example: car model with proportional plus integral feedback

Special choice

$$K_i = \frac{b}{m} K_p, \quad K(s) = K_p + \frac{K_i}{s} = K_p \frac{s + \frac{b}{m}}{s}$$

leads to a pole-zero cancellation

$$G(s) K(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}} \times K_p \frac{s + \frac{b}{m}}{s} = \frac{\frac{p}{m} K_p}{s},$$

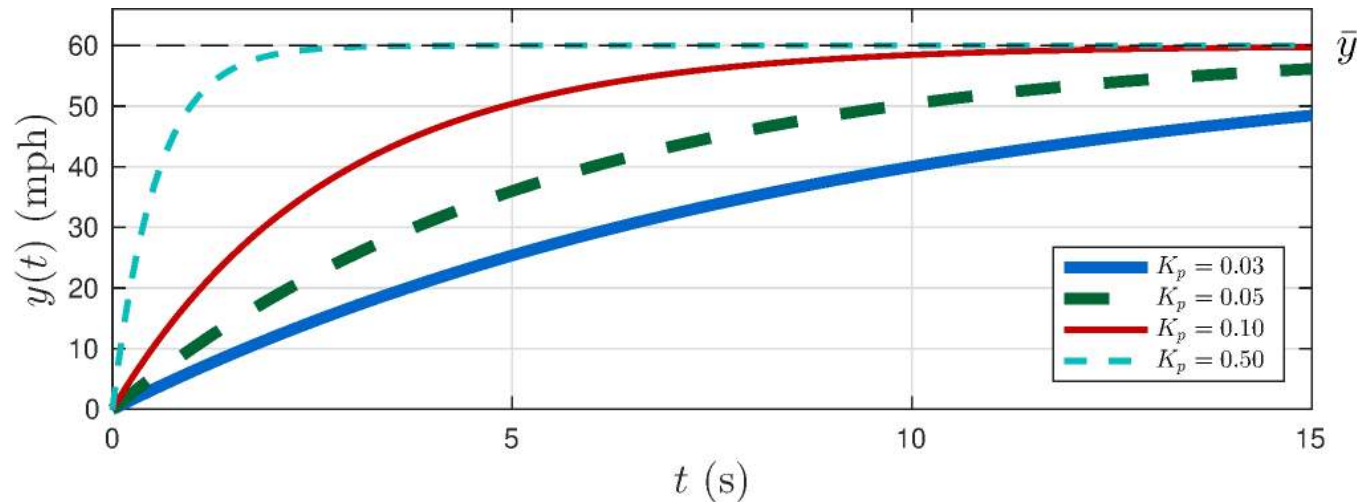
Closed-loop sensitivity transfer-functions

$$S(s) = \frac{1}{1 + G(s)K(s)} = \frac{s}{s + \frac{p}{m} K_p}$$

Zero at the origin: asymptotic tracking

No complex poles: no oscillations

Example: car model with proportional plus integral feedback



Asymptotic tracking, gone are the oscillations!

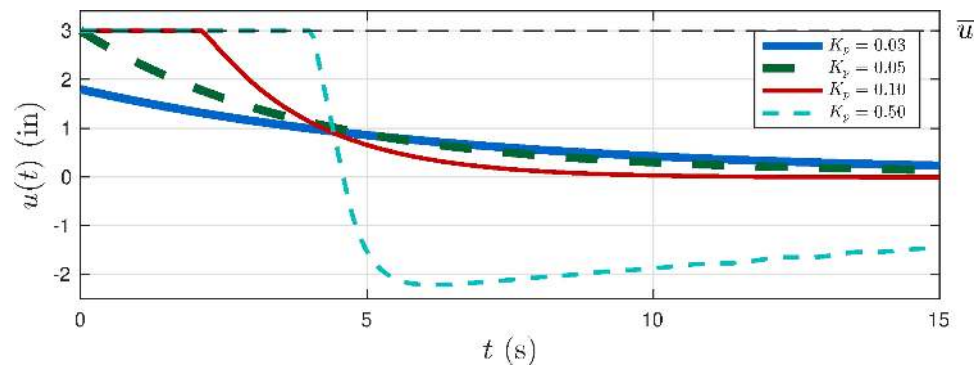
Beware of control effort: saturation occurs if

$$K_p > \frac{\bar{u}}{\bar{y}} = \frac{3}{60} = 0.05$$

4.3 Integrator Wind-up

Example: car model with proportional plus integral feedback

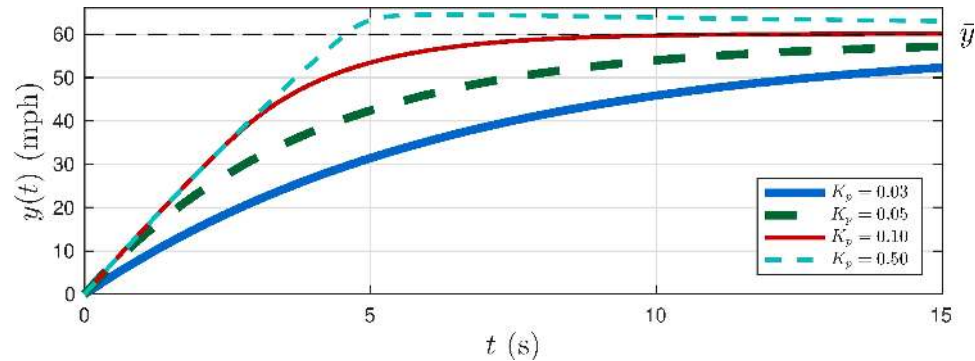
Nonlinear input saturation



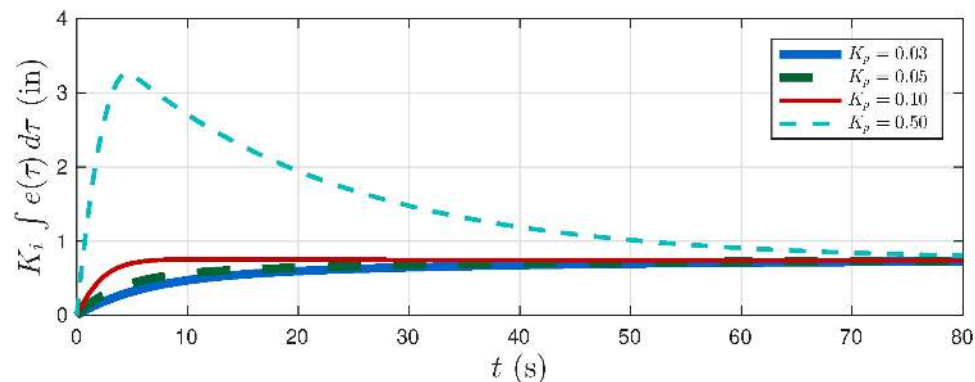
Output overshoots

Example: car model with proportional plus integral feedback

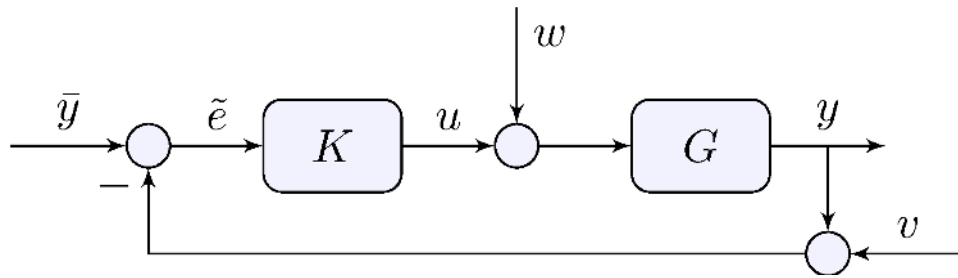
Output converges slowly



Integrator wind-up



4.4 Feedback with Disturbances



Loop relations

$$y = G(u + w), \quad u = K \tilde{e}, \quad \tilde{e} = \bar{y} - (y + v)$$

Closed-loop transfer-functions

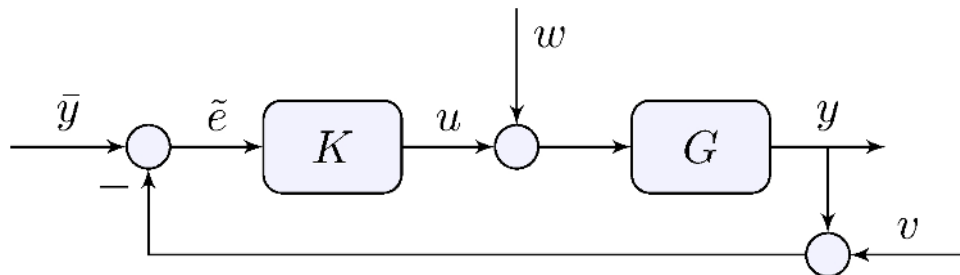
$$y = H \bar{y} - H v + D w, \quad u = Q \bar{y} - Q v - H w$$

Tracking error

$$e = \bar{y} - y = S \bar{y} + H v - D w$$

Beware $\tilde{e} \neq e$!

4.4 Feedback with Disturbances



Closed-loop transfer-functions

$$y = H \bar{y} - H v + D w$$

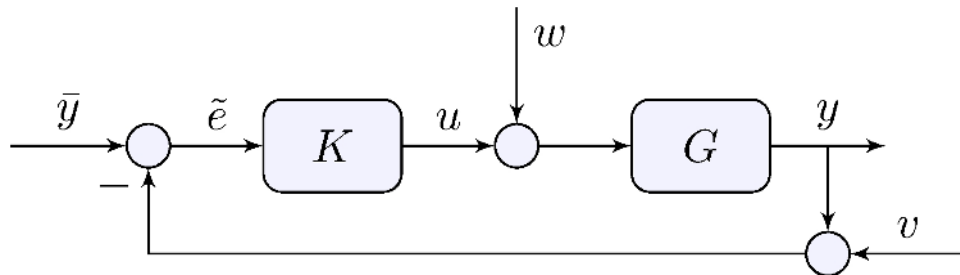
$$u = Q \bar{y} - Q v - H w$$

$$e = S \bar{y} + H v - D w$$

Gang of four

$$S = \frac{1}{1 + GK}, \quad D = \frac{G}{1 + GK}, \quad H = \frac{GK}{1 + GK}, \quad Q = \frac{K}{1 + GK}$$

4.4 Feedback with Disturbances

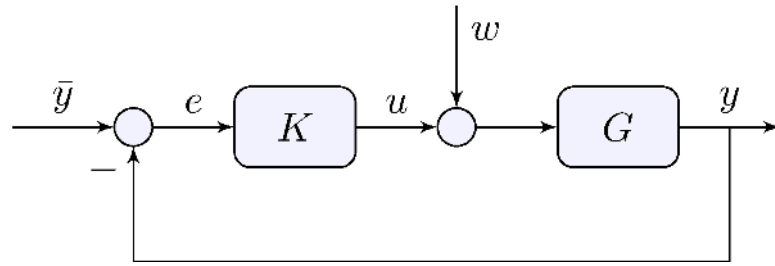


Gang of four

$$S = \frac{1}{1 + GK}, \quad D = GS, \quad H = GKS, \quad Q = KS$$

- S has as zeros the poles of GK
- No pole-zero cancellations in $GK \implies S, D, H, Q$ have the same poles!
- Stabilizing S stabilizes all other transfer-functions
- If there are pole-zero cancellations in GK be careful and see Section 4.7

4.5 Input-Disturbance Rejection



Closed-loop transfer-functions

$$e = S \bar{y} - D w, \quad D = GS = \frac{G}{1 + GK}$$

Lemma 4.2

Let $S = (1 + GK)^{-1}$ and $D = GS$ be asymptotically stable and

$$\bar{y}(t) = \bar{y} \cos(\omega t + \phi), \quad w(t) = \bar{w} \cos(\omega t + \psi)$$

If K has a pole at $s = j\omega$ then $\lim_{t \rightarrow \infty} e(t) = 0$

Proof: S and D have zeros at $s = j\omega$

Example: car model with proportional feedback

Closed-loop transfer-functions

$$S(s) = \frac{s + \frac{b}{m}}{s + \frac{b}{m} + \frac{p}{m}K}, \quad D(s) = S(s) G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m} + \frac{p}{m}K}$$

No asymptotic tracking, no asymptotic disturbance rejection

Example: car model with integral feedback

Closed-loop transfer-functions

$$S(s) = \frac{s \left(s + \frac{b}{m} \right)}{s^2 + \frac{b}{m}s + \frac{p}{m}K}, \quad D(s) = S(s) G(s) = \frac{\frac{p}{m}s}{s^2 + \frac{b}{m}s + \frac{p}{m}K}$$

Asymptotic tracking and asymptotic disturbance rejection, oscillations

Example: car model with integral feedback

Closed-loop transfer-functions

$$S(s) = \frac{s \left(s + \frac{b}{m} \right)}{s^2 + \frac{b}{m}s + \frac{p}{m}K}, \quad D(s) = S(s) G(s) = \frac{\frac{p}{m}s}{s^2 + \frac{b}{m}s + \frac{p}{m}K}$$

Asymptotic tracking and asymptotic disturbance rejection, oscillations

Example: car model with proportional plus integral feedback

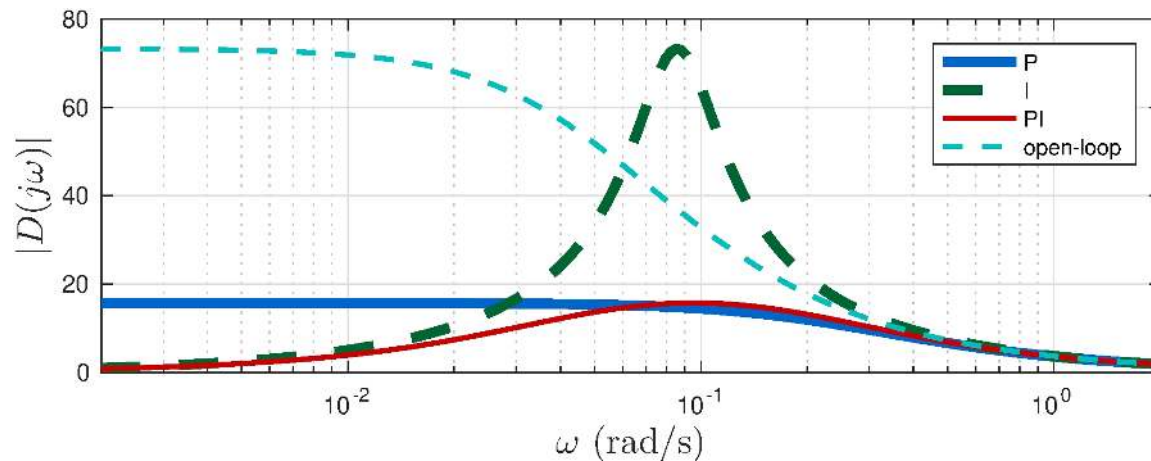
Closed-loop transfer-functions

$$S(s) = \frac{s}{s + \frac{p}{m}K_p}, \quad D(s) = S(s) G(s) = \frac{\frac{p}{m}s}{\left(s + \frac{b}{m} \right) \left(s + \frac{p}{m}K_p \right)}$$

Asymptotic tracking and asymptotic disturbance rejection, no oscillations

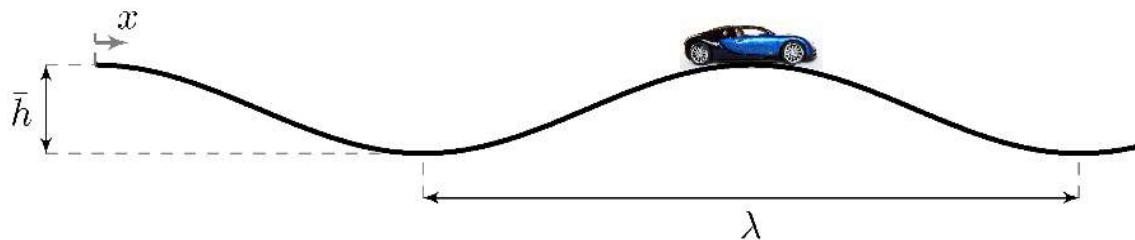
4.5 Input-Disturbance Rejection

Example: car model with various feedback controllers



Integral controller is sensitive to disturbances at 0.1 rad/s

Watch out for rolling hills!



Example: toilet bowl model

Loop relations

$$G(s) = \frac{1}{sA}, \quad K(s) = K$$

Closed-loop transfer-functions

$$S(s) = \frac{s}{s + \frac{K}{A}}, \quad D(s) = \frac{\frac{1}{A}}{s + \frac{K}{A}}$$

Asymptotic tracking but no asymptotic disturbance rejection!

Pole must be at the controller for asymptotic disturbance rejection

D does not have a zero at the origin

4.6 Measurement noise

Closed-loop transfer-functions

$$e = S \bar{y} + H v, \quad S = \frac{1}{1 + GK}, \quad H = GKS$$

Fundamental limitation

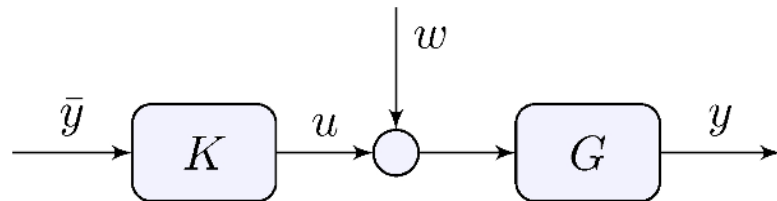
$$S + H = 1$$

At every frequency

$$S(j\omega) + H(j\omega) = 1$$

- Good tracking ($S(j\omega) \approx 0$) means bad measurement noise rejection ($H(j\omega) \approx 1$)
- Good measurement noise rejection ($H(j\omega) \approx 0$) means bad tracking ($S(j\omega) \approx 1$)
- Tracking requires good sensors!
- $|S(j\omega)| + |H(j\omega)| \neq 1$: both S and H can be large at the same time!

4.7 Pole-zero Cancellations and Stability



Open-loop

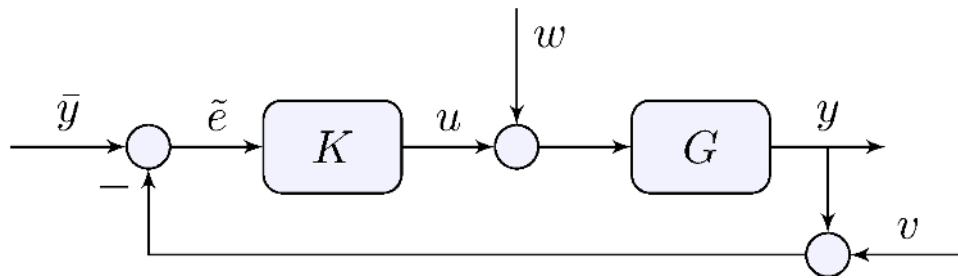
$$G = \frac{N_G}{D_G}, \quad K = G^{-1} = \frac{D_G}{N_G},$$

- G cannot be strictly proper or K will not be proper
- Even if G is proper but not strictly proper and $K = G^{-1}$

$$y = G K \bar{y} + G w = \bar{y} + G w, \quad u = K \bar{y}$$

- Both G and K must be stable or y or u will be unbounded
- Stabilization of unstable systems cannot be done in open-loop

4.7 Pole-zero Cancellations and Stability



Closed-loop

Internal stability if

$$S, \quad SG, \quad SK, \quad SGK$$

are all asymptotically stable

Lemma 4.3:

Internal stability if and only if S is asymptotically stable and any pole-zero cancellation in GK is of stable poles or zeros

Proof (sketch): If

$$G = \frac{(s - z)}{(s - p)} \tilde{G}, \quad K = \frac{(s - p)}{(s - z)} \tilde{K}, \quad GK = \tilde{G}\tilde{K}, \quad S = \tilde{S} = \frac{1}{1 + \tilde{G}\tilde{K}}$$

However

$$SG = \frac{(s - z)}{(s - p)} \tilde{S}\tilde{G}, \quad SK = \frac{(s - p)}{(s - z)} \tilde{K}\tilde{G}$$

so that z and p must both have negative real part

Example: car model with proportional plus integral feedback

Model and controller

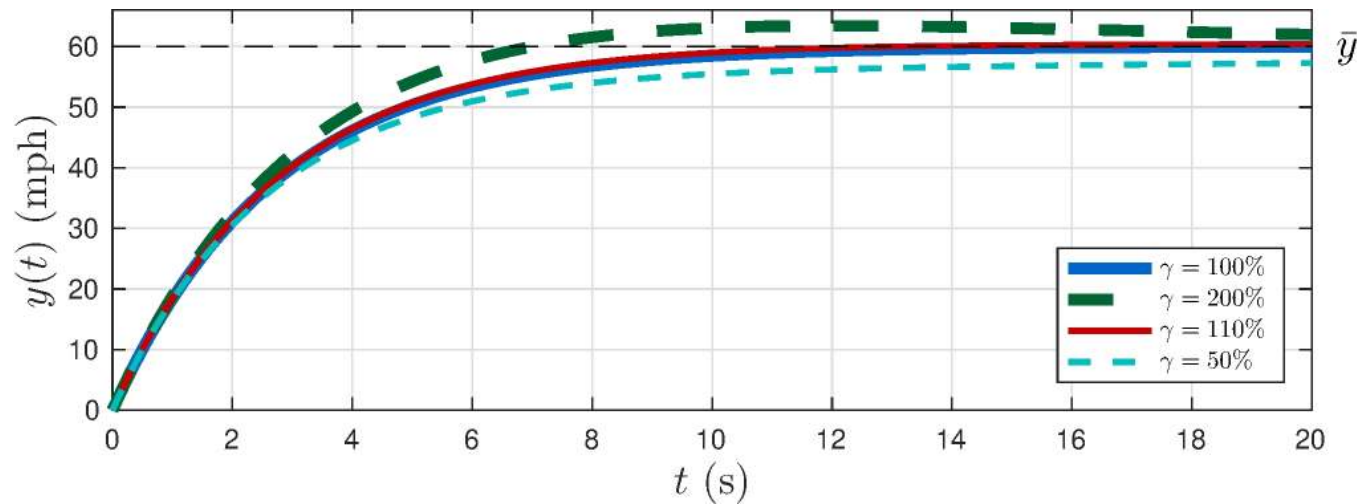
$$G(s) = \frac{p/m}{s + b/m}, \quad K(s) = K_p \frac{s + z}{s}, \quad z = \frac{K_i}{K_p} = \frac{b}{m}$$

Closed-loop transfer-functions

$$S(s) = \frac{s}{s + \frac{p}{m}K_p}, \quad H(s) = \frac{\frac{p}{m}K_p}{s + \frac{p}{m}K_p}, \quad Q(s) = \frac{K_p \left(s + \frac{b}{m}\right)}{s + \frac{p}{m}K_p}, \quad D(s) = \frac{\frac{p}{m}s}{\left(s + \frac{b}{m}\right) \left(s + \frac{p}{m}K_p\right)}$$

Canceled pole is a zero of Q and a pole of D

Example: car model with proportional plus integral feedback



Imperfect cancellation

$$K_i = \gamma \frac{b}{m} K_p$$

More to come in Chapter 6

Learn more

- [Book website](#)